

- Q.29 Write a note on (CO2)
 a) Compiler b) Interpreter
- Q.30 Explain if statement and differentiate between if and nested if statement. (CO2)
- Q.31 What are the limitations of arrays in C? (CO3)
- Q.32 Describe the use of the switch statement with an example. (CO3)
- Q.33 Explain the importance of input/output statements in C programming. (CO2)
- Q.34 Describe string handling in C and mention common string functions. (CO3)
- Q.35 List various data types available in C. (CO2)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Discuss the different types of control structures in C programming. Provide examples and explain how they are used to control the flow of a program. (CO2)
- Q.37 Write a program in C for finding the smallest element and the average of a single dimensional array. (CO3)
- Q.38 Write short notes on: (CO5)
 a) Software applications in Electronic Engineering

No. of Printed Pages : 4

202435/120935/030935/

Roll No.

106726/031333

3rd Semester/ Mechtronics Subject : Computer Programming & Applications

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is a valid declaration of a pointer in C? (CO1)
 a) Interptr; b) Intptr;
 c) Int ptr; d) ptr int;
- Q.2 What does the 'break' statement do in a loop? (CO2)
 a) Exits the current loop
 b) Continues to the next iteration
 c) Stops the program
 d) None of the above
- Q.3 Which of the following is the correct syntax to read a string from the user in C? (CO2)
 a) Scanf("%S", str); b) gets(str);
 c) input(str); d) Both (a) and (b)
- Q.4 In C, which operator is used to access the members of a structure? (CO2)
 a) .(dot) b) -> (arrow)
 c) :(colon) d) Both a & b

- Q.5 What will be the output of the following code snippet: `printf("%d", 5+3*2);` 2) (CO2)
 a) 11 b) 16
 c) 8 d) 10
- Q.6 Which function is used to dynamically allocate memory in C? (CO1)
 a) Malloc() b) alloc()
 c) new() d) Reserve()
- Q.7 In C, how do you declare an array of integers with 10 elements? (CO2)
 a) `int arr[10];` b) `int arr(10);`
 c) `array int arr[10];` d) `int arr {10};`
- Q.8 What is the result of the following expression: size of (char)? (CO2)
 a) 1 b) 2
 c) 4 d) Depends on the system
- Q.9 Which of the following control structures is used for conditional execution? (CO3)
 a) for loop b) if statement
 c) Switch statement d) Both b and c
- Q.10 The keyword used to define a constant value in C is: (CO2)
 a) Constant b) #define
 c) const d) Both b and c

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 The process of finding and fixing errors in a program is called _____. (CO1)

- Q.12 In C, the function used to get the length of a string is _____. (CO2)
- Q.13 An array can store multiple values of the same _____ type. (CO2)
- Q.14 Give one example of header file. (CO1)
- Q.15 List two relational operator. (CO2)
- Q.16 What is the use of flow chart? (CO1)
- Q.17 To write data to a file in C, the function used is _____. (CO3)
- Q.18 A loop that executes at least once before checking its condition is called a _____ loop. (CO3)
- Q.19 Alt + F9 is the shortcut key used for _____. (CO2)
- Q.20 Variable that can hold a reference to a memory address is known as a _____. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the concept of algorithms and their importance in programming. (CO1)
- Q.22 Describe the process of compiling and running a C program.
- Q.23 Describe the steps in development of a program. (CO1)
- Q.24 Draw a flow chart to find the sum of the marks of a student in his/her five subjects. (CO1)
- Q.25 Differentiate between while and do while loops. (CO2)
- Q.26 Differentiate between arrays and pointers. (CO2)
- Q.27 Differentiate call by reference and call by value. (CO2)
- Q.28 Explain break statement with suitable example. (CO2)